5. (a) The table shows information about the Haber process and the contact process.

	Haber process	Contact process
Raw materials	nitrogen comes from the air hydrogen is made from natural gas	sulfur is made from impurities in natural gas oxygen comes from the air
Conditions	iron catalyst 450 °C 200 atm	vanadium(V) oxide catalyst 450 °C 1 atm
Equation	$N_2 + 3H_2 \rightleftharpoons 2NH_3$	$2SO_2 + O_2 \rightleftharpoons 2SO_3$

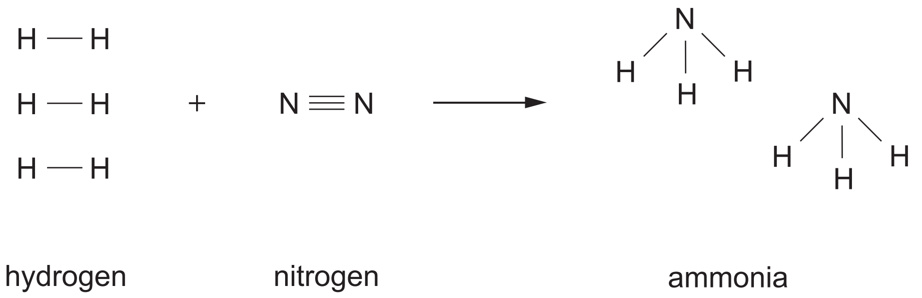
Use the information in the table and your own knowledge to answer this question.

State whether the statements below are **true** or **false**. [3]

	True or **false**?
Both processes use a catalyst that is a metallic element	....
Both processes are carried out at the same temperature and pressure	....
Both processes are reversible reactions	....
Both processes use air as a raw material	....

Examiner only

(b) The equation below shows the bonds that are broken and the bonds that are formed when ammonia is produced.



(i) The energy released when one N—H bond forms is 391 kJ.

Give the total number of N—H bonds in two molecules of ammonia and show that the amount of energy released when all the bonds are formed is 2346 kJ. [1]

(ii) The energy released when all the bonds in the **product** are formed is 2346 kJ.

The energy needed to break all the bonds in the **reactants** is 2253 kJ.

I. Give the term that describes a reaction where more energy is released in forming bonds than is needed to break the bonds in the reactants. [1]

....

II. Calculate the overall energy change that takes place during this reaction. [1]

Overall energy change = ... kJ

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(c) Ammonia is used to make ammonium sulfate and ammonium carbonate.

Tests can be carried out to identify the negative ion in each of these compounds.

Draw a line from each ion to the appropriate test and observation. [2]

Ion	**Test and observation**
	add dilute hydrochloric acid; gas formed turns limewater milky
	carry out a flame test; flame turns brick red
carbonate ion, CO_3^{2-}	
	add sodium hydroxide solution; pungent smelling gas is formed
sulfate ion, SO_4^{2-}	
	carry out a flame test; flame turns apple green
	add barium chloride solution; white precipitate is formed

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